

Rowzatul Zannath Prerona

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Bachelor of Science

Ahsanullah University of Science and Technology, Dhaka

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🐙 GitHub Profile

🌐 LinkedIn Profile

EDUCATION

- **Bachelor of Science in Computer Science and Engineering** 2022-26
Ahsanullah University of Science and Technology, Dhaka CGPA: 3.43
- **Higher Secondary Certificate** 2018-20
Holy Cross College, Dhaka GPA: 5.00
- **Secondary School Certificate** 2015-18
Ideal School and College, Motijheel, Dhaka GPA: 5.00

ACADEMIC PROJECTS

- **EfficientNet-CBAM: Attention-Guided Brain Tumor Classification**
An attention-guided deep learning framework for multi-class brain tumor classification from MRI scans.
 - Integrated Convolutional Block Attention Module (CBAM) into an EfficientNetB3 backbone to focus on pathologically relevant regions, achieving **99.29%** classification accuracy.
 - Implemented HiResCAM gradient-based visualizations for model explainability, improving clinical interpretability across Glioma, Meningioma, and Pituitary Tumor classes.
 - Technology Used: Python, TensorFlow, Keras, EfficientNetB3, CBAM, HiResCAM, OpenCV, NumPy, scikit-learn.
- **Deforestation Mapping in Dhaka — Semantic Segmentation**
Deep learning-based semantic segmentation to detect and monitor deforested areas using satellite imagery.
 - Built a high-resolution satellite imagery dataset of Dhaka spanning 2012–2022 and benchmarked three architectures: U-Net, Attention U-Net, and a hybrid ResNet50-Attention U-Net.
 - Evaluated models using Dice Score, IoU, F1-Score, Precision, and Recall; providing a reproducible framework for environmental change monitoring.
 - Technology Used: Python, TensorFlow, Keras, PyTorch, U-Net, ResNet50, NumPy, OpenCV, scikit-learn, matplotlib.
- **Planet Paw — Animal Rescue & Welfare Platform**
A multifaceted web platform integrating donation processing, pet adoption, medical aid listings, and food sales.
 - Designed a community-driven ecosystem supporting full CRUD operations for rescue listings, donor management, and inventory with emphasis on transparency and sustainability.
 - Built with a PHP/MySQL backend managed via XAMPP and phpMyAdmin, delivering seamless user engagement across multiple welfare services.
 - Technology Used: PHP, HTML, CSS, JavaScript, MySQL, XAMPP, phpMyAdmin.
- **LeafLedger — Bookshop Management System**
A full-stack bookshop management system streamlining inventory, sales, and customer operations.
 - Implemented book and genre management, shopping cart functionality, stock monitoring, order processing, and admin-level operations including book approvals and content management.
 - Built on ASP.NET Core MVC with a Bootstrap frontend and SQL Server database, providing a clean administrative interface suitable for real-world retail deployment.
 - Technology Used: ASP.NET Core MVC, Bootstrap, HTML, CSS, SQL Server, .NET 8 SDK, SSMS.
- **Spread Love for Pet — Pet Management Desktop Application**
A feature-rich Java desktop application consolidating pet management, health monitoring, and community engagement.
 - Developed user and pet profile modules, an expense tracker, a pet health encyclopedia, a BMI calculator, and a community connectivity module for pet owners.
 - Technology Used: Java.
- **Paw Prints Manager — Pet Care Mobile Application**
A comprehensive all-in-one pet care mobile application built with Flutter for pet owners.
 - Features include pet and user profile management, a monthly expense tracker, a health encyclopedia searchable by concern, a BMI calculator, and a reminder system for vet visits and medications.
 - Technology Used: Flutter, Dart.

PUBLICATIONS

•Spectral–Spatial Mamba with Uncertainty-Guided Refinement for Thyroid Nodule Diagnosis

2026

Array (Elsevier)

Published

- Co-authored with Mohammad Amanour Rahman; proposed SSMURNet, a spectral-spatial Mamba framework with evidential deep learning for confidence-calibrated thyroid nodule malignancy classification from ultrasound images, achieving 92.50% accuracy and 0.9700 AUROC.
- DOI: 10.1016/j.array.2026.101058

•Hard Spatial Gating for Precision-Driven Brain Metastasis Segmentation: Addressing the Over-Segmentation Paradox in Deep Attention Networks

2025

arXiv preprint, eess.IV / cs.CV

Preprint

- Sole-authored work introducing SG-Net, a precision-first spatial gating architecture that resolves the "over-segmentation paradox" in brain metastasis MRI segmentation, achieving a threefold improvement in boundary precision over Attention U-Net with $8.8\times$ fewer parameters.
- arXiv: 2511.22606

RESEARCH INTERESTS

Medical Image Processing, Medical Image Segmentation, Deep Learning for Radiology, Computer Vision, Uncertainty Quantification in Medical AI, Explainable AI (XAI) for Healthcare, AI-based Clinical Decision Support, Semantic Segmentation

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, HTML + CSS, Dart, Java

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Flutter, ReactJs, ASP.NET Core MVC, Bootstrap

ML/DL & CV Tools: OpenCV, NumPy, scikit-learn, matplotlib, CBAM, HiResCAM, EfficientNetB3, ResNet50, U-Net

Databases: MySQL, SQL Server, Firebase, MongoDB

Dev Tools: Git, GitHub, VS Code, XAMPP, phpMyAdmin, SSMS, .NET 8 SDK

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Database Management System, Operating Systems, Software Engineering

Soft Skills: Research-driven Problem Solving, Self-learning, Adaptability, Presentation